

Activity: Binary Search Using List in C++

Question

You are required to develop a **C++ program to demonstrate the Binary Search algorithm using a list.**

The program should create **four different sorted lists** containing **10, 100, 1,000, and 10,000 elements.**

Given Lists

List 1 – 10 Elements

10, 20, 30, 40, 50, 60, 70, 80, 90, 100

List 2 – 100 Elements

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ..., 98, 99, 100

List 3 – 1,000 Elements

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ..., 998, 999, 1000

List 4 – 10,000 Elements

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ..., 9998, 9999, 10000

Note: Students should use a loop to generate the 100, 1,000, and 10,000 element lists instead of manually typing all the values.

Requirements

Your program must:

1. Create the four lists using `list<int>`.
2. Ensure that all lists are arranged in **ascending order**.
3. Ask the user to enter a value to search.
4. Implement **Binary Search** to search for the value.
5. Display whether the value was found or not found.
6. Count and display the **number of comparisons** made.
7. Test the Binary Search algorithm using all four list sizes.
8. Display the results in a suitable format.

Sample Input

Enter value to search: 7350

Sample Output

===== BINARY SEARCH =====

List Size: 10
Search Value: 70
Result: Element Found
Comparisons: 3

List Size: 100
Search Value: 73
Result: Element Found
Comparisons: 6

List Size: 1000
Search Value: 735
Result: Element Found
Comparisons: 10

List Size: 10000
Search Value: 7350
Result: Element Found
Comparisons: 13